

Instructor: Date:

Learner: Regional Branch:

Program: Center:

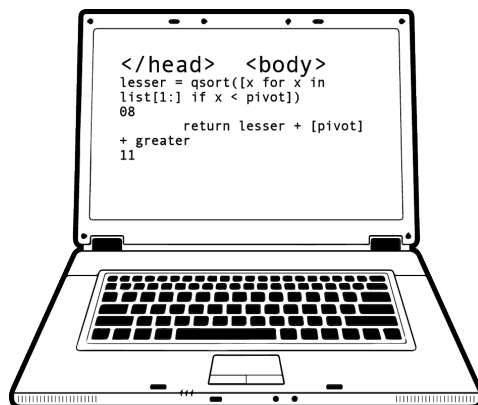
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

This is how the server works in ASP.NET.



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can list the properties of the server object in ASP.NET.
- I can recall some methods of the server object in ASP.NET.
- I can talk about properties and methods of the server object in ASP.NET.

Ask your partner:

- What do you know about the page life cycle in ASP.NET?
- Do you know some of the properties of the page class in ASP.NET? Which ones?





Let's Explore!

Thomas is giving a lecture about the Server Object in ASP.NET.



Instructions: Read the text below.

Thomas: Hi, everybody! I'm Thomas, the person who is going to be in charge of today's lecture. During this presentation, I will try to explain the most relevant aspects of the Server Object that we use in ASP.NET. First of all, it is quite important to know that this object is an instance of the `System.Web.HttpServerUtility` class, and it is used to provide functionalities to the server through different properties and methods. But what do we know about the Server Object properties? Well, there are mainly two properties that this object offers. The first property, which is known as `MachineName`, is the one you use to get the name of a local computer. The second one is `ScriptTimeout` which we use for getting and setting the amount of time of a request.

Concerning methods, there are several that are defined by the Server Object. Among them, `Server.CreateObject`, `Server.Execute`, `Server.GetLastError`, `Server.HtmlEncode`, `Server.MapPath`, `Server.Transfer` and `Server.UrlEncode`. But what do all these methods do? `CreateObject` generates an instance of a server component or COM; `Execute` runs .asp files; `GetLastError` describes errors conditions, and `HtmlEncode` turns a specified string into HTML encoding.

Furthermore, we have `MapPath`, which is the method that we can use when we need to map a specified virtual file path on the server. There is also `Transfer`, which sends current state information to another web page, and `URL encode` which applies URL encoding to a specified string.

Finally, you must note that all these methods and properties are only run through the intrinsic Server Object that is provided by ASP.NET.



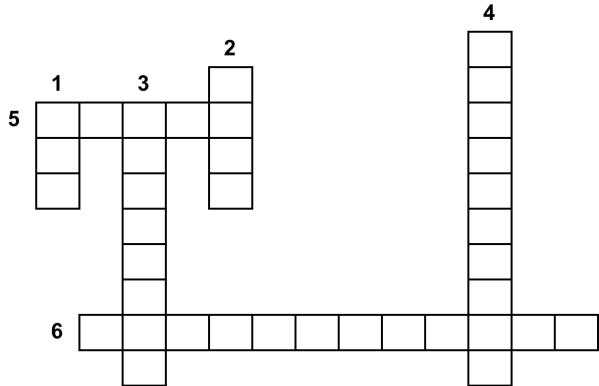
Let's Practice!

What do we know about the Server Object properties?

Instructions: Read the clues. Complete the crossword.

Down

- Hi, everybody! I'm Thomas, the person _ _ _ is going to be in charge of today's lecture about the server object in ASP.NET.
- During this presentation, I will try to explain the most relevant aspects of the Server Object _ _ _ **t** we use in ASP.NET.
- First of all, it is quite important to know that this object is an _ _ _ _ _ _ _ **e** of the System.Web.HttpServerUtility class.
- There are mainly two **p** _ _ _ _ _ _ _ _ _ **s** that the Server Object offers.



Across

- The first property, _ _ _ _ **h** is known as MachineName, is the one you use to get the name of a local computer.
- The second one is **s** _ _ _ _ _ _ _ _ _ _ **t** which we use for getting and setting the amount of time of a request.



Let's Practice!

Thomas is explaining some Server Object Methods.

Instructions: Read the text. Circle the correct words to complete the text.

There are several methods that (*is defining / is defined / are defined*) by the Server Object in ASP.NET, they (*is used / are using / are used*) to perform different actions during a user session, among them, we have, Server.CreateObject, Server.Execute, Server.GetLastError, and Server.HtmlEncode.

Now, I would like to mention some of the actions that (*executed / is executed / are executed*) by these methods, for example, CreateObject (*know / is known / are known*) for being able to generate instances of server components; and Execute (*is considered / are considered / considered*) a key element because it allows running .asp files.





Extension Activity

Making an Explanatory Chart About the Server Object in ASP.NET.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the methods and properties of the Server Object in ASP.NET.
- Identify key ideas in the information you read.
- Make a chart to explain some properties and some methods with your own words.
- Prepare an oral presentation to explain your chart.
- Share your work with the class.

Self-Assessment

Yes

No

- I look for more information about the methods and properties of the Server Object in ASP.NET.
- I identify key ideas in the information I read.
- I make a chart to explain some properties and methods with my own words.
- I prepare an oral presentation to explain my chart.
- I share my work with the class.

☐	☐
☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- ScriptTimeout** (noun): a property that can get and set in seconds the time that a script needs to run before its time is up
e.g. The second one is ScriptTimeout which we use for getting and setting the amount of time of a request.
- server component** (noun): a server control that is used in ASP.NET pages in the user's interface
e.g. CreateObject generates an instance of a server component or COM.
- string** (noun): a sequence of data that includes characters, spaces, and numbers
e.g. HTML Encode turns a specified string into HTML encoding.

- virtual file path** (noun): a file that is visible in a browser window
e.g. Furthermore, we have MapPath, which is the method that we can use when we need to map a specified virtual file path on the server.
- state information** (noun): all the current values of a user session
e.g. There is also Transfer, which sends current state information to another web page, and URL encode which applies URL encoding to a specified string.



Learning Points

Relative Clauses: Object Pronouns.

You use **relative clauses** to provide extra information about a subject or an object mentioned in a previous clause.

To give additional information about an object, we need to use **object relative pronouns** such as **who**, **whom**, **which**, and **that**. We use **who**, **whom**, and **that** for people; and **which** or **that** for things.

It's important not to confuse the subject and object pronouns in the relative clauses; when the relative pronoun is used as a **subject pronoun**, the main verb goes after it; but when the relative pronoun is used as an **object pronoun**, there is always a noun or a pronoun after it.

Examples:

- *This is the manual **that** covers everything about ASP.NET. (Subject relative pronoun)*
- *This is the manual **that** he gave me, it covers everything about ASP.NET. (Object relative pronoun)*

Another way to identify **object relative pronouns** is the fact that they can be omitted from the sentence without changing its meaning:

- *This is the manual about ASP.NET **that** he gave me.*
- *This is the manual about ASP.NET he gave me.*

There are two types of clauses: **defining and non-defining**. In the **defining** relative clauses, we add information that is significant to the previous clause while in the **non-defining** ones the extra information is not relevant.

Let's see the grammatical structures below:

Defining Relative Clauses

Subject + **object pronoun** + noun or pronoun + complement.

Examples:

- *The page life cycle **which** we use to contain various controls is instantiated as a control object.*
- *She is the student **whom** I met in programming class.*
- *The app **that** he created is very interesting.*





Learning Points

Non-Defining Relative Clauses

Subject, + **object pronoun** + noun or pronoun+ complement, + other clause.

Examples:

- *The professor, **who** we met yesterday, recommended us a very useful manual.*
- *The manual, **which** he gave me, explained all about the server object.*
- *The page class, **which** you use, has extremely useful properties.*

Examples

- *I will try to explain the most relevant aspects of the Server Object that we use in ASP.NET.*
- *First of all, it is quite important to know that this object is an instance of the System.Web.HttpServerUtility class.*
- *The second one is ScriptTimeout which we use for getting and setting the amount of time of a request.*

Some Methods in the Server Object in ASP.NET.

The following chart provides information about some of the most important methods used by the Server Object in ASP.NET:

Method	Functions
CreateObject (String) CreateObject (type)	Creates an instance of a class by its ProID. Creates an instance of a class according to its type.
Equals (Object) Execute (String)	Validates if an object is equal to the current one. Starts the handler for the virtual path once a request is made.
Execute (String, Boolean) GetLastError GetType	Clears the QueryString and Form Collections. Gives back a previous error. Shows the Type of an instance.
HtmlEncode HtmlDecode ToString	Changes a string to HTML characters. Changes HTML characters to ordinary ones. Gives back a string for the current object.
Transfer (String) UrlDecode UrlEncodeToken	Finishes processes in one page and starts them on another. Changes URL strings to ordinary. Changes strings by using Base64-encoded data.